



Omnia

The Omnia Protocol

Explained — From First Principles

A plain-language introduction for people
who know nothing about Omnia.

Replacing trust with mathematics

Public-domain (CC0) · Mid-2026 edition

A plain-language introduction — no background required

This guide assumes you know nothing about blockchains, cryptography, or “web3.” It starts from ordinary ideas and builds up, one step at a time, to what the Omnia Protocol actually is and why it was built. Every technical claim here comes from Omnia's own public code and documentation, including the parts that are still unfinished — Omnia's own rule is “no marketing metrics,” and this guide keeps to it.

SECTION 1

Start with a plain question

Imagine a notebook that the whole world shares. Anyone can write in it, nobody can secretly erase a line, and everyone can check that what's written is true — without having to trust the person who wrote it, or any company in the middle. That shared, tamper-proof notebook is the idea behind Omnia.

The catch is making that notebook **fast**, **fair**, and **truly owned by its users** rather than by a corporation. That is the problem Omnia sets out to solve.

In plain terms

Omnia is a shared record of truth that no single person or company controls. You don't have to trust anyone — you can check everything yourself, because the rules are math, not promises.

SECTION 2

What is a “protocol,” really?

A **protocol** is just a set of rules that any computer can follow to talk to any other. Email is a protocol. The web (HTTP) is a protocol. Nobody “owns” email — Gmail, Outlook and Apple Mail all speak the same rules, so they work together.

Omnia is a protocol in exactly this sense. In its own words:

Omnia's own definition

“Omnia is not a company, a coin, or an app. It is a protocol — a fundamental set of rules that any computer can follow to participate in a shared, unchangeable record of truth.”

That distinction matters. Apps and coins come and go; a protocol is infrastructure. Omnia is released into the public domain (a CC0 license), so anyone can build on it freely.

SECTION 3

The problems it's trying to fix

Omnia starts from a list of things it believes are broken in today's digital systems, and pairs each with its approach:

The problem	Omnia's approach
Blockchains are slow and expensive, wasting energy	Process transactions in parallel instead of one-at-a-time
Online governance is opaque; votes get ignored	Quadratic voting with reputation that decays over time
Companies profit from your personal data	You control your data; prove facts with zero-knowledge proofs
Crypto rewards speculation and hoarding	"Universal Basic Compute" given to every participant

You don't need to understand each row yet — the rest of this guide unpacks the ideas behind them.

SECTION 4

The big idea: replace trust with math

Most systems ask you to **trust** someone: a bank, a platform, a "verified" badge. Omnia's core move is to replace trust with **proof**. Instead of "trust us, this is correct," every claim on the network comes with a cryptographic proof that anyone can check by running a simple calculation.

If the numbers add up, the claim is true. If they don't, it's rejected. There is no authority to appeal to and none to bribe — verification is the same for everyone, everywhere.

Why this is powerful

"No oracles, no authorities, no 'trust us.'" Checking whether something is valid is a pure calculation on the data itself — so two strangers who don't trust each other can still agree on what happened.

SECTION 5

A different way to agree: the causal graph

Traditional blockchains write history as a single line — one block after another, in strict order. That's simple, but it's a bottleneck: everyone waits their turn, like a single checkout lane at a shop.

Omnia instead records events in a **causal graph**. Picture a web of events where each new event points back to the ones it depends on. Two events that have nothing to do with each other (say, a payment in Tokyo and a vote in Berlin) don't have to wait for one another — they can be recorded at the same time. Only events that genuinely depend on each other are ordered.

In plain terms

Instead of one checkout lane, Omnia opens many lanes at once. Unrelated things happen in parallel; only things that actually affect each other are kept in order. The technical name is “causal graph consensus” (a DAG with vector clocks and CRDTs) — a way for many computers to agree on history without a single line and without a central referee.

This is what lets Omnia aim for high throughput while staying decentralized.

SECTION 6

How fast is “final”? Two lanes

When you send something, you want to know it's **done** — settled, irreversible. Omnia gives you two stages so you get a fast answer and a rock-solid one:

- **Lane 0 — instant pre-confirmation.** Within a moment, a group of validators cryptographically sign off that they've seen your event. You get near-instant feedback.
- **Lane 1 — canonical finality.** The event is woven permanently into the shared history, and later settled for good.

The Omnia Wallet shows this honestly: a completed transfer is labelled “Final · Lane 0” so you always know which stage you're looking at. This two-lane design is specified in Omnia's ADR-025 and checked with formal (mathematical) models on every code change.

SECTION 7

Who are you? Identity you actually own

On most apps, your account belongs to the company: they can lock it, lose it, or mine it. Omnia flips this with **self-sovereign identity**.

Your identity is a **DID** (a “decentralized identifier,” written like `did:omnia:...`). It is derived from a secret key that is generated and stored **only on your device**. No company holds it. You prove who you are by signing a short challenge with your key — the key itself never leaves your phone.

What this means for you

You hold the keys, literally. Nobody can freeze or seize your identity. The trade-off is real responsibility: if you lose your device and your recovery phrase, no one can restore it for you — so you back it up. Omnia also supports social recovery (trusted guardians) to soften that.

SECTION 8

UBC: Universal Basic Compute

Every registered identity receives a monthly allowance of **Universal Basic Compute (UBC)** — a built-in budget for using the network. Two things make it unusual:

- **It's given to everyone**, as a quota that refreshes each epoch (period) — not something you buy or mine.
- **It's “soulbound.”** It is bound to your identity and cannot be traded away or accumulated into a fortune.

Important: UBC is not money

UBC is a utility credit for *using compute* on the network — think of it like a monthly data allowance, not a currency. It can't be bought, it isn't an investment, and “sending” it spends it rather than enriching a recipient. This design deliberately avoids the speculation and hoarding common in crypto.

SECTION 9

Settlement-agnostic: not an island

Omnia doesn't demand that the whole world move onto one new chain. It is **settlement-agnostic**: it can anchor its proofs onto existing networks — Ethereum today, and Bitcoin, Solana or others as adapters mature — wherever there's data availability and proof verification.

In plain terms

Omnia can plug into the financial and blockchain systems that already exist, rather than trying to replace them. It settles where it makes sense to settle.

SECTION 10

How it's built: five layers

Under the hood, Omnia is organized as a stack, each layer resting on the one below — much like how the internet stacks simple layers into something powerful:

Layer	What it does
5 — Economics	UBC allowance and governance (voting)
4 — Identity	Your DID, key recovery, agent & biometric anchors
3 — Binding	Provenance — proving where things came from
2 — Domain shards	Specialized areas that scale independently
1 — Substrate	The causal graph itself — the shared record
0 — Settlement	Anchoring proofs to outside chains (ZK-rollup)

SECTION 11

Is it real yet? An honest status

Omnia's team publishes its progress plainly, including what is *not* finished. As of mid-2026:

- **Live testnet.** A public node runs in production, with a multi-node validator mesh reaching real BFT finality (in one stress run, 10,000 of 10,000 events finalized across all five validators). It has been proven across a three-region internet network, not just one machine.
- **Mobile wallet shipped.** The Omnia Wallet (v1) runs against that live node: create a wallet, log in with your on-device key, receive a 1,000-UBC monthly quota, send, and view history.
- **Measured, not marketed.** Performance figures come from reproducible benchmarks on stated hardware (e.g. a consensus pipeline around 12,000 ops/sec; finality latency in microseconds on a single node).
- **Still stubs.** Some pieces are honestly marked unfinished — several settlement adapters (Bitcoin, Solana, Cosmos), proof-of-useful-work, radio-fingerprinting hardware, and a broader validator network are not production-ready yet.

The honest bottom line

Omnia is a working, live testnet with a real wallet and formally-checked consensus — not a whitepaper, and not a finished mainnet. It is a serious system that is candid about the road still ahead.

SECTION 12

Mini-glossary

Protocol	A shared set of rules any computer can follow — like email or the web. Not a company or a product.
Causal graph	Omnia's way of recording events as a web of dependencies, so unrelated things can happen in parallel.
Consensus	How independent computers come to agree on the same history without a central boss.
Validator	A computer that helps check and finalize events on the network.
Finality	The point at which a transaction is settled and can't be reversed. Omnia has two stages: Lane 0 (instant) and Lane 1 (canonical).
DID	A decentralized identifier (did:omnia:...) derived from a key on your own device — an identity you control.
Self-custody	You, and only you, hold the keys to your identity and assets. No company can freeze them.
UBC	Universal Basic Compute — a monthly, soulbound allowance for using the network. A utility credit, not money.
Soulbound	Permanently tied to one identity; can't be traded or transferred away.
Zero-knowledge proof	A way to prove a fact (e.g. 'I'm over 18') without revealing the underlying data.
Settlement-agnostic	Able to anchor its records onto different external blockchains, rather than being locked to one.

SECTION 13

Where to go next

If you'd like to see it for yourself:

- **Try the live node:** <https://78.47.43.136.sslip.io/api/v1/node/info>
- **Website:** the Omnia Protocol site (About, Architecture, FAQ, and the wallet's privacy & terms).
- **Wallet:** the Omnia Wallet app — self-custody identity and your UBC, on your phone.
- **Code & community:** the open-source repositories on GitHub and the community Discord.

This document is a plain-language introduction compiled from Omnia Protocol's public code and documentation. The protocol is released into the public domain (CC0). Figures and status reflect the project as of mid-2026 and will evolve.